SQLChange: An Agentic Reasoning Pipeline for SQL Query Optimization

Ahmed Khan Fahim Islam Dev Rathod Arnav Akula Devarchith Parashara Batchu

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Problem & Motivation

Problem

- SQL optimization is manual and requires expertise
- Performance bottlenecks are hard to identify before execution
- Existing tools lack reliable verification of correctness and speedups

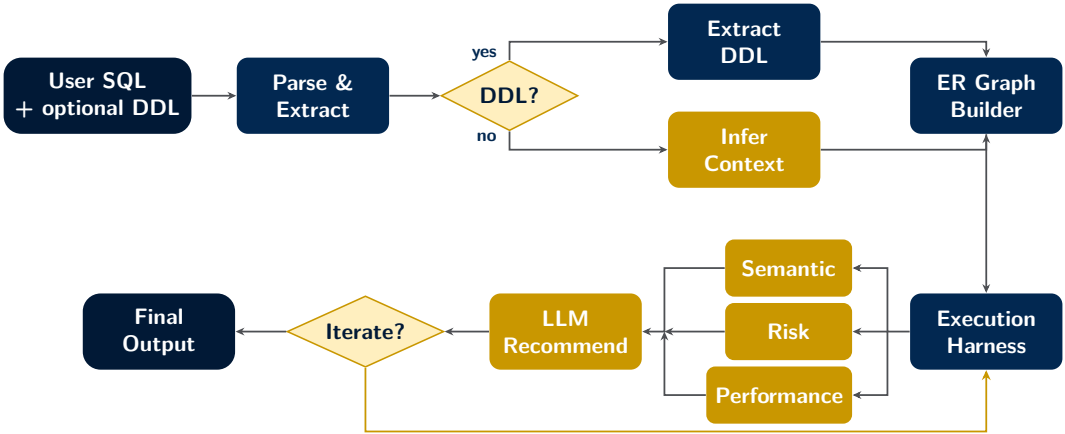
Motivation

- Petabyte-scale data makes inefficient queries expensive
- Privacy constraints limit the use of external AI tools
- Engineers need an end-to-end agent for optimization, validation, and explanation

Our Question

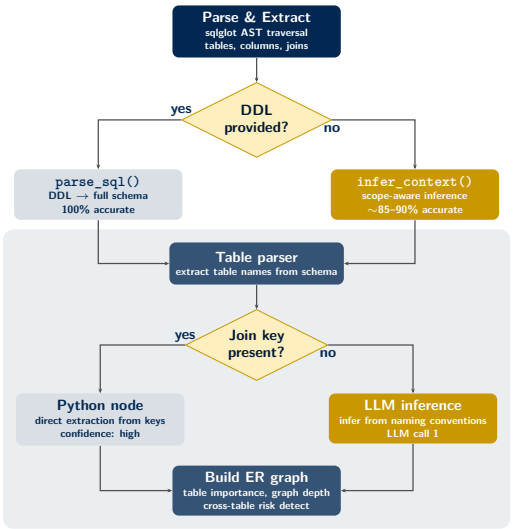
Can an AI agent automatically optimize SQL queries while preserving **correctness** and improving **performance**?

Agentic Reasoning Pipeline



DDL = Data Definition Language (`CREATE TABLE`, schema definitions)

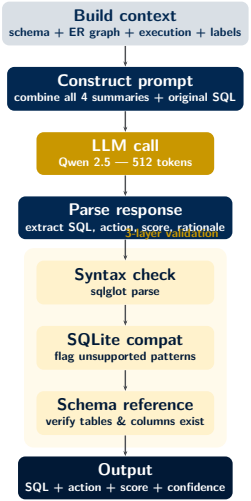
Parsing & ER Graph Construction



ER Graph Construction

- **Table importance hierarchy:**
  - *Root* — never a join target (changes cascade everywhere)
  - *Intermediate* — both source and target
  - *Leaf* — only a target (safest to modify)
- **Cross-table risk signals:** graph depth, join-WHERE overlap, dependency fan-out
- Feeds context to recommendation engine and risk labeler downstream

Recommendation Engine



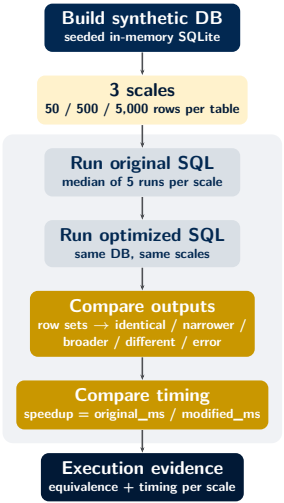
LLM-Powered Optimization

- LLM receives **full upstream context**: schema summary, ER graph hierarchy, execution evidence, and prior iteration labels
- Outputs: optimized SQL, action type (`optimize` / `rewrite` / `keep_original`), confidence (1–10), rationale
- On later iterations: sees why the last attempt scored poorly and adjusts (conservative if risk ≥ 7)

Three-Layer Validation

- **Syntax** — sqlglot must parse output as valid SQL
- **SQLite compat** — flags DATE_SUB, INTERVAL, IF(), ILIKE, RIGHT JOIN; suggests equivalents
- **Schema ref** — every table/column must exist in schema; validation fails → confidence downgraded

Execution Harness & Labeling



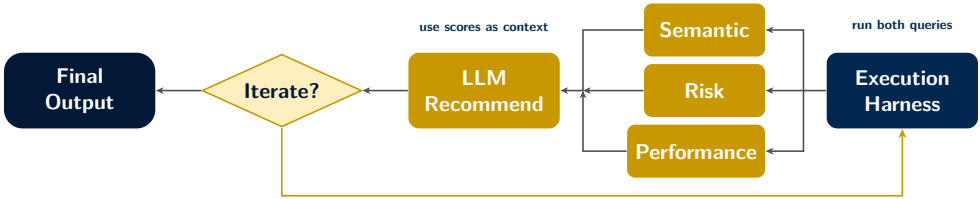
Synthetic Database

- Deterministic: seeded by query hash, reproducible across runs
- 75% join match rate with realistic boundary values for WHERE clauses
- SQL normalized for SQLite (DATE_SUB → strftime, true → 1)

Reasoning Labels

- **Semantic** — deterministic classification from execution output (identical / narrower / broader / different); LLM refines only when confidence is low
- **Performance** (1–10) — multi-scale timing; 8–10 = major speedup, 1–2 = degradation
- **Risk** (1–10) — table importance hierarchy, join depth, cross-table WHERE, row-count mismatches

Iterative Refinement



How It Works

- Harness executes original + optimized SQL on synthetic data at 3 scales
- Labelers score: deterministic rules first, LLM only if confidence is low
- Scores feed back to LLM — it sees *why* the last attempt failed
- High risk → conservative mode; wrong results → prioritize correctness

Stop Conditions

- Perf ≥ 8 **and** risk ≤ 3
- LLM recommends `keep_original`
- SQL fails validation
- Max iterations reached (default: 3)

Evaluation

Evaluation Setup

- **100 queries** built from gretelai/synthetic_text_to_sql, excluding development set
- Stratified: 34 simple, 37 moderate, 29 complex
- **20 degraded queries** — programmatically injected inefficiencies (unnecessary subqueries, redundant joins)
- **20 no-DDL queries** — schema must be inferred
- Each query run through full pipeline with 5-minute timeout, up to 3 iterations

Metrics

- **Semantic equivalence** — does the optimized query return the same results?
- **Performance score** (1–10) — speedup across 3 scales
- **Risk score** (1–10) — safety of the change
- **Validity** — does the recommendation parse and match the schema?
- **Iteration depth** — passes before convergence
- **Wall-clock time** — end-to-end latency

Results & Analysis

Complexity Tier	Queries	Sem. Equiv.	Avg. Perf.	Avg. Risk	Avg. Iters
Simple (basic SQL, single join)	34	79%	6.6	3.6	1.2
Moderate (aggregation, subqueries)	37	73%	6.8	4.0	1.7
Complex (multi-join, window, set)	29	52%	7.2	3.8	1.8
Overall (100 queries)	100	68%	6.9	3.8	1.6

Key Findings

- **Reliable:** only 1 parse error in 100 queries
- **Conservative:** 55% kept original — avoids unnecessary optimization. Optimization rate scales with complexity (12% simple → 51% moderate → 66% complex)
- **Anti-pattern detection:** 75% of degraded queries correctly fixed — SELECT * and unnecessary subqueries reliably caught
- **DDL-robust:** equivalence similar for DDL vs. no-DDL — inference path performs just as well

Limitations

- **Complex queries:** only 52% equivalence — 14 of 22 non-equivalent results were complex queries; joins and window functions cause subtle result-set changes
- **Iteration penalty:** single-iteration queries scored 7.5 avg perf vs. 5.6 for multi-iteration — harder queries that need more passes are also harder to optimize well
- **Risk is conservative:** only 15% of optimizations flagged high-risk, appropriate for production SQL

Conclusions & Future Work

What We Built

- End-to-end agentic pipeline that *generates* optimized SQL — the proposal planned comparison only; we went further
- Execution-verified: synthetic DB testing catches silent result-set changes that LLM-only approaches miss
- Iterative feedback loop — labels feed back to the LLM for progressive improvement
- Schema inference from queries alone — works without DDL, which the proposal assumed would always be available
- Runs fully local via Ollama — no data leaves the machine. Evaluation on Qwen2.5:7b

Limitations & Future Work

- SQLite only — future: multi-dialect support (PostgreSQL, MySQL)
- 110 queries vs. the proposed 250–300 pairs — future: Spider/BIRD benchmarks and real GitHub commit pairs
- No baseline comparison — future: zero-shot LLM vs. structured prompt
- Scales limited to 5k rows — future: real database connections for production-scale testing
- Local 7B model — future: larger or fine-tuned models for better output
- RAG for query-plan-aware optimization (original reach goal)

Thank You

Questions?